

EXPERT KNOWS BEST? LEVERAGING MOE ROUTING SIGNALS FOR KV CACHE EVICTION: PITFALLS AND LESSONS LEARNED

Anonymous authors

Paper under double-blind review

ABSTRACT

Mixture-of-Experts (MoE) Transformers achieve high model capacity with sparse computation, yet their KV caches remain fully dense, creating a memory bottleneck. We investigate whether MoE expert routing decisions—computed for free during the forward pass—can serve as zero-overhead importance signals for KV cache eviction. Our hypothesis is that tokens consistently routed to specialized experts (low routing entropy) carry higher task-relevant content and should be preferentially retained. We propose **RouteKV**, a training-free eviction framework that uses cross-layer routing entropy as an importance score. Our investigation reveals a critical pitfall: in standard MoE training with load-balancing objectives, routing entropy is nearly uniform across all tokens (gap ≈ 0.02 – 0.03 nats), rendering the routing signal uninformative for eviction and causing RouteKV to perform worse than random on several benchmarks. However, when explicit routing specialization supervision is introduced, entropy gaps exceed 0.93 nats and RouteKV achieves perfect key-token recall even at 25% KV budget. Crucially, routing-based importance rankings are nearly orthogonal to attention-based rankings (Spearman $r \approx 0$), suggesting that routing provides a *complementary* rather than redundant signal—but only when routers are explicitly trained to specialize.

1 INTRODUCTION

The KV cache is a dominant memory bottleneck in large language model inference, growing linearly with sequence length. For MoE models such as Mixtral (?) and DeepSeek-V2 (?), this bottleneck is compounded: while computation is sparse (only a subset of experts process each token), the KV cache remains fully dense, storing keys and values for every token regardless of semantic importance.

Existing KV cache eviction methods—H2O (?), StreamingLLM (?), SAGE-KV (?), CAKE (?), NACL (?)—rely on attention score statistics or key similarity to rank token importance. These methods are architecture-agnostic and ignore a rich structural signal unique to MoE models: the expert routing decision for each token, which is already computed at zero marginal cost during the forward pass. We investigate a natural hypothesis: *tokens routed to specialized experts (low routing entropy) are more semantically important and can guide KV cache eviction.*

This paper documents the journey from this appealing hypothesis to its practical pitfalls. Our contributions are: (1) We identify that standard MoE training produces near-uniform routing entropy, making routing-based eviction unreliable—RouteKV performs comparably to or worse than random eviction. (2) We show that with explicit routing specialization supervision, entropy gaps exceed 0.93 nats, enabling perfect key-token recall at 25% KV budget. (3) We establish that routing signals are nearly orthogonal to attention scores (Spearman $r \approx 0.002$ – 0.020), suggesting a complementary rather than redundant signal. (4) We provide ablations over loss functions, expert counts, aggregation strategies, and layer windows to characterize when routing specialization emerges and how to best exploit it.

2 RELATED WORK

KV cache eviction has attracted substantial attention as a means of reducing memory during long-context inference. H2O (?) identifies “heavy hitter” tokens via accumulated attention scores. StreamingLLM (?) retains attention sink tokens and recent tokens. SAGE-KV (?) performs one-time top- k selection post-prefill. CAKE (?) models token importance as a cascading, layer-adaptive process. NACL (?) combines proxy-token eviction with random eviction to mitigate attention bias. KeyDiff (?) uses key similarity rather than attention scores. All these methods are architecture-agnostic and discard MoE-specific routing information. On the MoE side, the sparse gating mechanism of ? forms the basis of modern MoE architectures including Mixtral (?), DeepSeek-V2 (?), and DeepSeek-V3 (?). These models use top- k routing with load-balancing auxiliary losses that, as we show, actively suppress the routing specialization needed for eviction. To our knowledge, no prior work exploits routing logits as a KV cache importance signal.

3 METHOD

For a token t at position i , let $p_\ell(t) \in \Delta^{E-1}$ denote the softmax routing distribution over E experts at MoE layer ℓ . The routing entropy is $H_\ell(t) = -\sum_e p_\ell(t)_e \log p_\ell(t)_e$. The RouteKV specialization score aggregates across layers: $S(t) = -\frac{1}{L} \sum_{\ell=1}^L H_\ell(t)$, where higher $S(t)$ (lower entropy) indicates more specialized routing. Tokens are ranked by $S(t)$ and the top- k are retained with no additional computation beyond the standard MoE forward pass.

Standard MoE training includes a load-balancing auxiliary loss that encourages uniform token dispatch (?), directly suppressing the routing entropy gap between semantically important and redundant tokens. To validate the hypothesis under favorable conditions, we introduce an explicit specialization loss: key tokens are pushed toward a designated expert via cross-entropy on routing logits, while filler tokens are pushed elsewhere. The combined objective is $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{spec}}$ with $\lambda = 0.5$.

4 EXPERIMENTS

We evaluate on a controlled MoE Transformer (4 layers, 4 heads, $d = 128$, 4 experts, top-1 hard routing) trained on three datasets: a synthetic 4-class classification task, SST-2 sentiment classification, and AG News topic classification. Sequences have length 48 with the first 9 positions designated as key tokens. We compare **RouteKV** (routing entropy), **Attention** (accumulated attention scores), and **Random** eviction at retain ratios $\{0.25, 0.40, 0.60, 0.75\}$. For the baseline phase, we use soft top-2 routing with 8 experts and standard load-balancing; for the research phase, we use hard top-1 routing with 4 experts and the specialization loss. Full hyperparameters are in Appendix G.

Pitfall 1: Standard routing provides no useful eviction signal. With standard load-balanced training, attention-based eviction achieves 88–94% KV Retention Accuracy (KVRA) at 40% budget, while RouteKV achieves only 43–71% KVRA—often worse than random (52–67%). The routing entropy gap is only 0.019–0.026 nats, confirming that load-balancing actively destroys the specialization signal needed for eviction (Figure 3, Appendix A).

Pitfall 2: Routing entropy is orthogonal to attention, not a proxy for it. As shown in Figure 1C, routing specialization scores and attention-based importance rankings have near-zero Spearman correlation across all datasets (synthetic: $r = 0.0016$; SST-2: $r = 0.0204^*$; AG News: $r = -0.007$, where the asterisk denotes marginal significance at $p < 0.05$). This orthogonality means routing is not a cheaper proxy for attention—it captures a fundamentally different dimension of token importance, suggesting potential value in hybrid scoring but ruling out routing as a drop-in replacement.

Routing specialization enables effective eviction. With explicit specialization supervision, entropy gaps grow to 1.05/1.01/0.93 nats for synthetic/SST-2/AG News respectively (Figure 1A–B), far exceeding the 0.05 nats specialization threshold. Key tokens exhibit mean routing entropy of 0.07–0.22 nats while filler tokens reach 1.12–1.16 nats—a 5–10 $\times$ difference. Figure 2 shows that RouteKV then matches attention-based eviction exactly, both maintaining 1.0 accuracy down to 25% KV budget, while random eviction degrades to 0.95–0.97 at the same budget. The consistency of

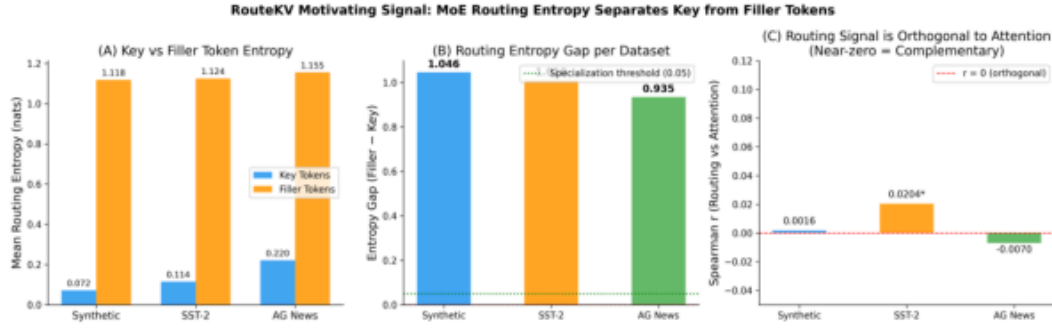


Figure 1: After explicit routing specialization training on a small controlled MoE (4 layers, $d = 128$, 4 experts): (A) key tokens have dramatically lower routing entropy than filler tokens (0.07–0.22 vs. 1.12–1.16 nats), a 5–10 $\times$ difference; (B) entropy gaps (0.93–1.05 nats) far exceed the 0.05 nats specialization threshold; (C) routing scores are nearly orthogonal to attention scores (Spearman $r \approx 0$), indicating complementarity rather than redundancy. Asterisk on SST-2 in (C) denotes marginal significance ($p < 0.05$).

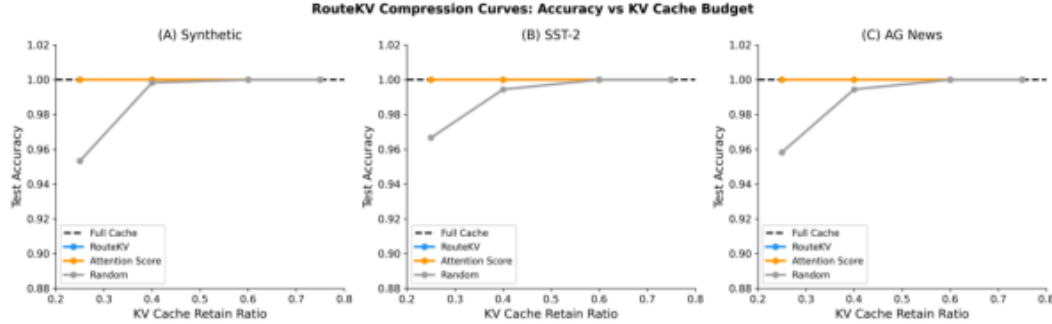


Figure 2: Compression curves (accuracy vs. KV retain ratio) after specialization training across three datasets. RouteKV and attention-based eviction maintain near-perfect accuracy across all budgets, while random eviction degrades at 25% retain ratio (0.95–0.97 accuracy). The Full Cache baseline is shown as a dashed reference. The near-perfect overlap of RouteKV and Attention curves across all three datasets confirms that routing entropy—when properly specialized—is as effective as attention-score-based eviction.

this result across all three datasets confirms that routing entropy, when properly specialized, provides a signal as discriminative as accumulated attention scores. RouteKV also achieves perfect key-token recall (1.0) at all retain ratios from 0.25 to 0.75, with precision decreasing from ≈ 0.75 at 25% to ≈ 0.25 at 75% as expected, since with perfect recall, precision = $n_{\text{key}}/n_{\text{retained}}$ (Figure 4, Appendix B).

Loss function ablation. Table 1 summarizes the effect of different routing objectives. Cross-entropy specialization achieves the highest entropy gap (1.02–1.05 nats) with perfect eviction accuracy and recall at 40% budget. Contrastive margin loss achieves moderate gaps (≈ 0.43 nats) while still attaining perfect eviction accuracy, suggesting that even partial specialization is sufficient once a minimum threshold is exceeded. Crucially, standard load-balancing produces near-zero or negative entropy gaps and catastrophic eviction accuracy (0.40–0.63 relative accuracy), confirming that the auxiliary loss used in production MoE models is the primary practical obstacle to routing-guided eviction. Notably, the No Routing Loss condition produces slightly negative entropy gaps, meaning key tokens exhibit marginally *higher* entropy than filler tokens—the reverse of the desired behavior. Full per-variant heatmaps are provided in Figure 6 (Appendix D).

Table 1: Loss function ablation: entropy gap and RouteKV performance at 40% KV budget (mean over three dataset variants). Cross-entropy specialization is critical; standard load-balancing destroys the routing signal.

Loss Type	Entropy Gap (nats)	Rel. Accuracy @ 40%	Key Recall @ 40%
Cross-Entropy Spec.	~ 1.03	1.000	1.000
Contrastive Margin	~ 0.43	1.000	1.000
No Routing Loss	~ 0.00	~ 0.48	~ 0.39
Load Balance	~ 0.00	~ 0.51	~ 0.30

5 CONCLUSION

We investigated MoE routing signals as zero-overhead importance cues for KV cache eviction. The central pitfall is clear: standard load-balanced MoE training produces near-uniform routing entropy, making routing-guided eviction unreliable and sometimes worse than random. The hypothesis only holds when routers are explicitly trained to specialize, at which point routing achieves perfect key-token recall even at 25% KV budget. Furthermore, routing importance rankings are nearly orthogonal to attention scores, suggesting a complementary rather than redundant signal. Key lessons for the community: (1) do not assume off-the-shelf MoE routers encode token importance; (2) load-balancing losses actively suppress specialization; (3) if routing specialization can be encouraged via modified training objectives, routing provides a genuinely complementary signal to attention. Future work should explore hybrid routing-attention eviction, online routing-guided eviction during decoding, and whether fine-tuned MoE models exhibit stronger routing specialization without explicit supervision.

REFERENCES

SUPPLEMENTARY MATERIAL

A BASELINE RESULTS: ROUTEKV WITHOUT SPECIALIZATION TRAINING

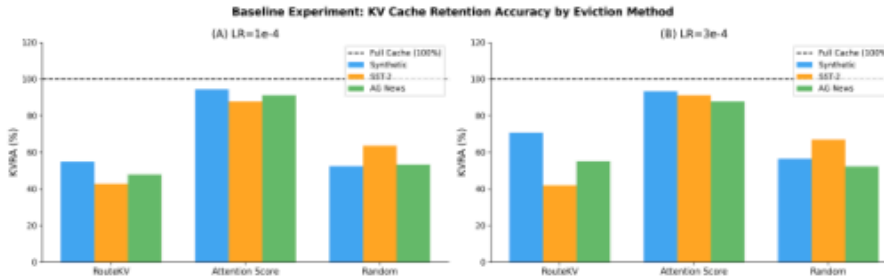


Figure 3: Baseline experiment (standard load-balanced training, soft top-2 routing, 8 experts): KV Retention Accuracy at 40% budget for two learning rates ($LR \in \{10^{-4}, 3 \times 10^{-4}\}$). Attention-based eviction achieves 88–94% KVRA; RouteKV achieves only 43–71%, often below random eviction (52–67%). The routing entropy gap is only 0.019–0.026 nats, confirming that load-balancing suppresses the specialization signal. These results motivate the need for explicit specialization supervision.

B KEY-TOKEN PRECISION AND RECALL

C TRAINING DYNAMICS

Figure 5 shows training and validation loss/accuracy curves for all three datasets under the research experiment with specialization loss. All datasets converge to near-perfect accuracy within 5–10

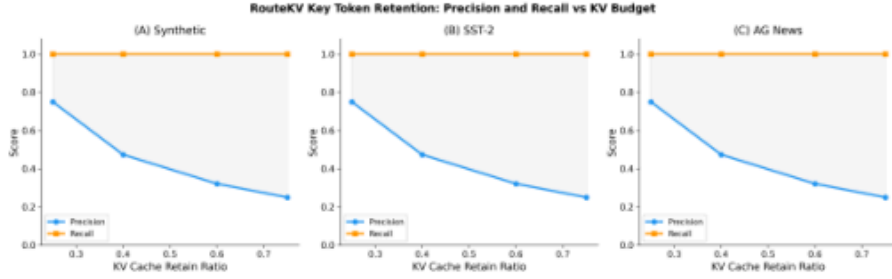


Figure 4: RouteKV key-token precision and recall vs. KV cache retain ratio (after specialization training). Recall is perfect (1.0) at all budgets across all three datasets. Precision decreases from ≈ 0.75 at 25% retain to ≈ 0.25 at 75% retain (slightly lower for SST-2 at 25%) as more non-key tokens are included at larger budgets—mathematically expected since with perfect recall, precision $= n_{\text{key}}/n_{\text{retained}}$. The consistency across datasets confirms that routing entropy reliably identifies key tokens after specialization training.

epochs. SST-2 exhibits a transient validation spike around epoch 25 (validation loss reaching ~ 2.0 , accuracy dropping to ~ 0.6 – 0.7), marked by a dashed vertical line; this is attributed to a transient routing collapse where the specialization loss temporarily disrupts the learned routing assignment before recovering.

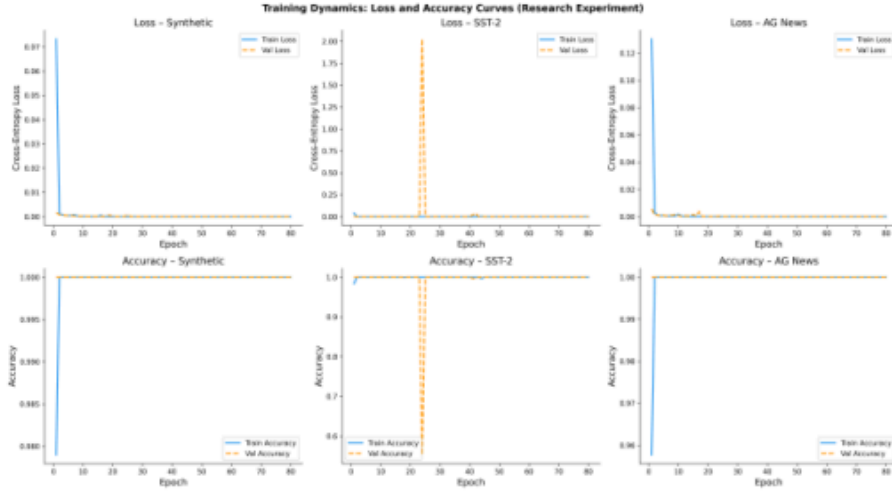


Figure 5: Training and validation loss/accuracy curves for all three datasets (research experiment with specialization loss). All datasets converge to near-perfect accuracy within 5–10 epochs. The SST-2 transient validation spike at epoch ≈ 25 (dashed vertical line) is attributed to a routing collapse where the specialization loss temporarily disrupts routing before recovery. Synthetic and AG News show no such anomalies.

D LOSS FUNCTION ABLATION: PER-VARIANT HEATMAPS

Figure 6 shows per-variant heatmaps of routing entropy gap, eviction accuracy, and key-token recall across loss function types. The x-axis variants are labeled Front-loaded, Scattered, and High-class, corresponding to structured key placement, mixed/scattered key placement, and high-class-count difficulty levels respectively. Cross-entropy specialization consistently produces the largest entropy gaps and perfect recall. Notably, the No Routing Loss condition produces slightly negative entropy gaps (key tokens have marginally higher entropy than filler tokens), while Load Balance achieves only 0.102 key-token recall on the High-class variant—the worst result across all conditions.

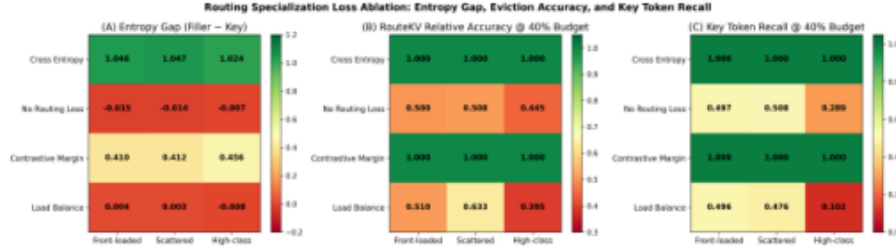


Figure 6: Heatmaps of (A) routing entropy gap, (B) RouteKV relative accuracy at 40% budget, and (C) key-token recall at 40% budget, across loss function types (rows) and dataset variants (columns: Front-loaded = structured key placement, Scattered = mixed placement, High-class = high class count). Cross-entropy specialization achieves entropy gaps of 1.02–1.05 nats and perfect recall. Contrastive margin achieves perfect accuracy and recall despite lower entropy gaps (0.41–0.46 nats). No Routing Loss produces slightly negative entropy gaps (key entropy exceeds filler entropy). Load-balancing achieves only 0.102 recall on the High-class variant—the worst result across all conditions.

E SCORE PLACEMENT AND NORMALIZATION ABLATIONS

Figure 7 shows that routing specialization is broadly distributed across MoE layers: all placement configurations (All Layers, Early Only, Late Only, Middle Only) achieve entropy gaps of 0.8–1.0 nats with modest differences, and all maintain near-perfect eviction accuracy. Figure 8 shows that normalization strategy matters substantially: the default ($T = 1.0$) is robust, but rank normalization causes catastrophic failure (near-zero accuracy and recall for all dataset variants), and high temperature ($T = 2.0$) substantially reduces entropy gaps (≈ 0.6 nats) and causes accuracy collapse for the multiclass variant. These results highlight that while the choice of layer window is relatively unimportant, the score normalization strategy can make or break the eviction pipeline.

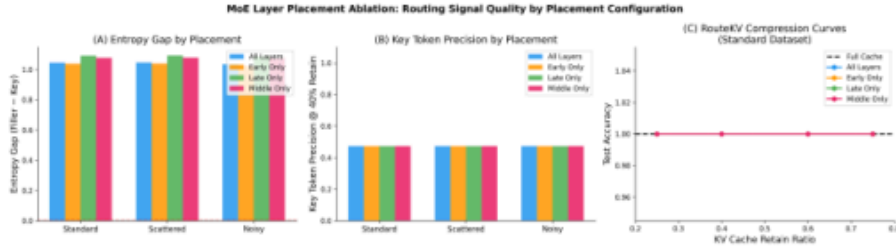


Figure 7: MoE layer placement ablation (All Layers, Early Only, Late Only, Middle Only): (A) entropy gap across dataset types—all configurations achieve 0.8–1.0 nats with modest differences; (B) key-token precision at 40% retain ratio; (C) compression curves on the standard dataset showing all placement strategies maintain near-perfect accuracy. Routing specialization is broadly distributed across layers, and the choice of layer window has limited impact on eviction quality.

F ADDITIONAL ABLATIONS

Number of experts. Figure 9 shows the effect of expert count on routing specialization. For the noisy dataset variant, the entropy gap increases monotonically with expert count (from ≈ 0 at 2 experts to ≈ 2.5 at 16 experts). However, standard, sparse, and dense variants show little increase, remaining near zero—indicating the entropy gap trend is driven primarily by the noisy variant and does not generalize across all dataset types. Key-token recall reaches 1.0 at 4 or more experts for the noisy variant; other variants achieve high recall (≈ 0.8 – 1.0) even at 2 experts, suggesting that more experts are only beneficial when the routing signal is difficult to learn. Importantly, the noisy variant achieves *higher* key-token precision (≈ 0.45 – 0.50) than other variants at the same budget, reflecting that the noisy variant’s routing specialization is sufficiently strong to preferentially select key tokens even at low retain ratios.

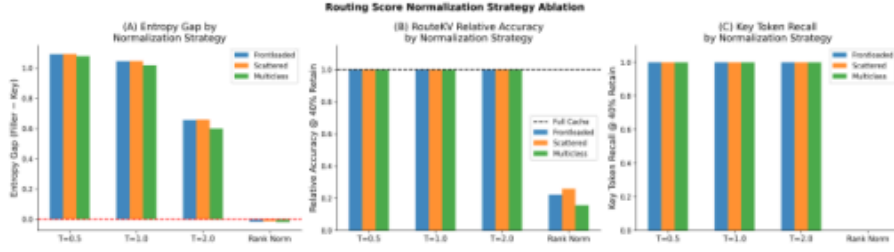


Figure 8: Routing score normalization ablation (temperature $T \in \{0.5, 1.0, 2.0\}$ and rank normalization): (A) entropy gap by strategy—high temperature ($T = 2.0$) reduces gaps to ≈ 0.6 nats; rank normalization collapses entropy gaps to near zero; (B) RouteKV relative accuracy at 40% retain—rank normalization causes catastrophic failure (near-zero accuracy for all variants); $T = 2.0$ causes accuracy collapse for the multiclass (High-class) variant; (C) key-token recall at 40% retain showing the same failure pattern. The default formulation ($T = 1.0$) is robust; both aggressive temperature scaling and rank normalization can be severely harmful.

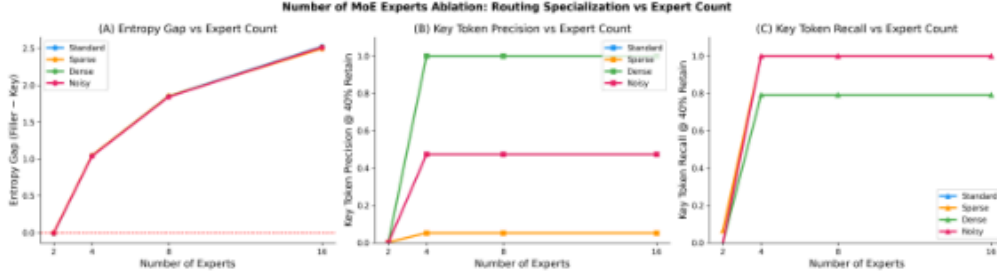


Figure 9: Number of experts ablation: (A) entropy gap vs. expert count—only the noisy dataset variant shows a monotonic increase (from ≈ 0 to ≈ 2.5 nats); other variants remain near zero. (B) key-token precision at 40% retain ratio by dataset variant—the noisy variant achieves higher precision (≈ 0.45 – 0.50) than other variants, reflecting strong routing specialization. (C) key-token recall at 40% retain ratio—the noisy variant reliably reaches 1.0 at 4 or more experts, while other variants achieve high recall even at 2 experts. More experts enable greater routing specialization primarily in high-noise settings.

Score aggregation strategy. Figure 10 compares five strategies for aggregating routing entropy across layers (Mean All, Last Layer, First Layer, Max Entropy, Weighted Last). Last-layer and weighted-last aggregation yield the highest entropy gaps (≈ 1.05 and 1.04 nats respectively), while first-layer aggregation is weakest (≈ 0.94 nats). Note that the y-axis in panel A starts at ≈ 0.2 nats rather than zero, which visually amplifies the apparent differences; in absolute terms, the range (0.94 – 1.05 nats) is narrow and all strategies substantially outperform attention-based and random baselines on the sparse dataset variant. Panel B confirms clear key-vs-filler entropy separation across all strategies on the standard dataset.

Layer aggregation window. Figure 11 shows that later layers carry stronger routing specialization signals. On the high-noise dataset (panel B), using only the last layer yields the most specialized key-token routing ($H \approx 0.09$ nats) compared to all-layer averaging ($H \approx 0.27$ nats) and early-layers-only ($H \approx 0.40$ nats, the least specialized), while filler token entropy remains consistently high (≈ 1.1 – 1.2 nats) across all strategies. Panel A shows entropy gaps of 0.8 – 1.1 nats across strategies and dataset types. Precision at 40% budget is largely similar (≈ 0.45 – 0.47) for standard and high-noise datasets across all strategies, while the long-key dataset achieves near-perfect precision (≈ 0.9 – 1.0) regardless of layer window.

Positional encoding. Figure 12 shows that positional encoding type has minimal effect on routing specialization quality. Entropy gaps are 0.88 – 1.07 nats across all PE variants and datasets (panel A). Panel B shows key-token routing entropy increasing from synthetic (≈ 0.05 – 0.10 nats) to AG News

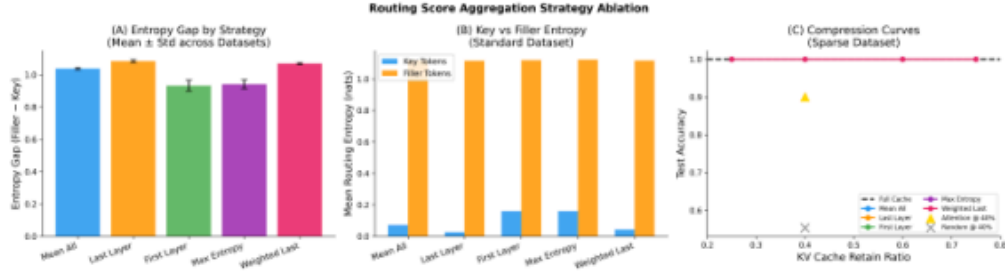


Figure 10: Score aggregation strategy ablation (Mean All, Last Layer, First Layer, Max Entropy, Weighted Last): (A) entropy gap by strategy (mean $\pm$ std across datasets, y-axis truncated at ≈ 0.2 nats)—Last Layer and Weighted Last achieve the highest gaps (≈ 1.05 and 1.04 nats), First Layer is weakest (≈ 0.94 nats); differences are narrow in absolute terms; (B) key vs. filler entropy on the standard dataset showing clear separation across all strategies; (C) compression curves on the sparse dataset showing all aggregation strategies substantially outperform attention and random baselines.

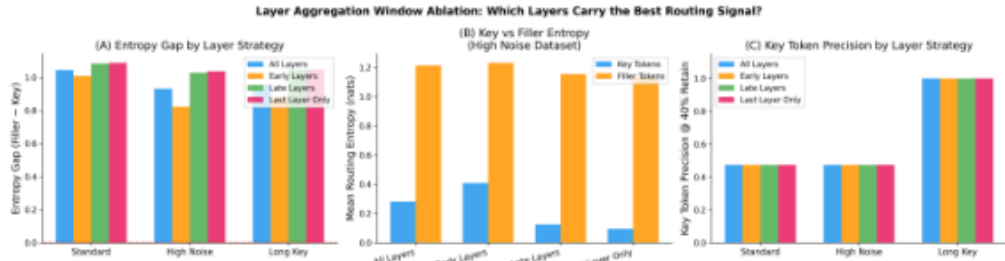


Figure 11: Layer aggregation window ablation (All Layers, Early Layers, Late Layers, Last Layer Only): (A) entropy gap across dataset types—all strategies achieve 0.8 – 1.1 nats with modest differences; (B) key vs. filler entropy on the high-noise dataset—later layers yield lower key-token entropy ($H \approx 0.09$ nats for last-layer-only vs. $H \approx 0.27$ nats for all-layer averaging and $H \approx 0.40$ nats for early-layers-only), while filler entropy remains high (≈ 1.1 – 1.2 nats); (C) key-token precision at 40% retain ratio—standard and high-noise datasets achieve ≈ 0.45 – 0.47 across all strategies; the long-key dataset achieves near-perfect precision (≈ 0.9 – 1.0).

(≈ 0.20 – 0.25 nats), with learned PE yielding slightly higher key-token entropy (less specialized), particularly for AG News. Key-token precision at 40% budget is nearly identical (≈ 0.47) across all conditions (panel C), confirming robustness to PE choice.

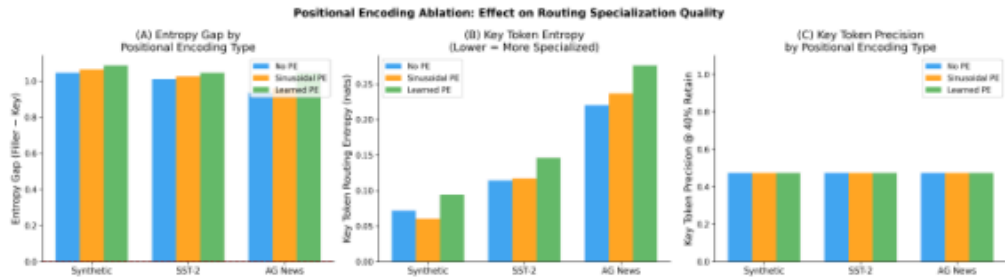


Figure 12: Positional encoding ablation (No PE, Sinusoidal PE, Learned PE): (A) entropy gap across datasets—all variants achieve 0.88 – 1.07 nats; (B) key-token routing entropy—learned PE yields slightly higher entropy (less specialized), particularly on AG News; (C) key-token precision at 40% retain ratio—all conditions achieve ≈ 0.47 , confirming robustness to positional encoding choice.

G EXPERIMENTAL HYPERPARAMETERS

Table 2 summarizes the key hyperparameters used in the research (specialization) experiments. The baseline experiments use soft top-2 routing with 8 experts and a standard load-balancing auxiliary loss coefficient of 0.01, trained with Adam at learning rates $\{10^{-4}, 3 \times 10^{-4}\}$ for 50 epochs.

Table 2: Hyperparameters for the main research experiments (specialization training).

Hyperparameter	Value
Layers	4
Attention heads	4
Hidden dimension d	128
Number of experts	4 (default; ablated over $\{2, 4, 8, 16\}$)
Routing type	Hard top-1
Sequence length	48
Key token positions	First 9 positions
Specialization loss weight λ	0.5
Optimizer	Adam
Learning rate	10^{-3}
Batch size	64
Training epochs	80
KV retain ratios evaluated	$\{0.25, 0.40, 0.60, 0.75\}$
Positional encoding (default)	Sinusoidal
Score aggregation (default)	Mean across all MoE layers